Arquitectura de Robots

Inverse Kinematic

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Section 1.1

Solvability and Existence of Solutions

The Inverse Kinematics Problem

1 Inverse Manipulator Kinematics

Core Challenge

Given end-effector pose ${}^0_N T$, solve for joint angles $\theta_1, \theta_2, \ldots, \theta_n$ in nonlinear equations:

$$ {}^0_N T = f(\theta_1, \theta_2, \ldots, \theta_n) $$

Case Study: PUMA 560

- 16 equations (12 non-trivial) for 6 unknowns
- 3 independent rotation equations + 3 position equations
- Nonlinear transcendental equations

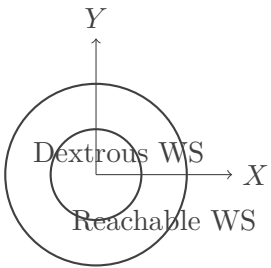
$$ \begin{cases} \text{Rotation matrix constraints} & (3 \text{ independent}) \\ \text{Position vector constraints} & (3 \text{ equations}) \end{cases} $$

Key Considerations in Solvability

1 Inverse Manipulator Kinematics

Existence of Solutions

- Goal must lie within workspace
- Two workspace definitions:
  - Dextrous WS: All orientations
  - Reachable WS: 1 orientation



Workspace Factors

- Joint limits (e.g., $\theta_2 \in [0°, 180°]$)
- Link lengths ($l_1 \neq l_2$ effects)
- Tool-frame transformation

Workspace Analysis: 2-Link Planar Manipulator

1 Inverse Manipulator Kinematics

$$\begin{cases} \text{Reachable WS} = \begin{cases} \text{Disc of radius } 2l_1 & \text{if } l_1 = l_2 \\ \text{Annulus } |l_1 - l_2| \leq r \leq l_1 + l_2 & \text{otherwise} \end{cases} \\ \text{Dextrous WS} = \begin{cases} \text{Origin} & \text{if } l_1 = l_2 \\ \emptyset & \text{otherwise} \end{cases} \end{cases}$$

Example

For $l_1 = l_2 = 1$ m:

- Boundary orientations: 1 solution
- Interior points: 2 solutions
- Workspace area $= 4\pi$ m^2

Degrees of Freedom Implications

Fundamental Limitation

Manipulators with <6 DOF cannot reach arbitrary poses in 3D space

- 4-5 DOF arms: Restricted motion subspace
- Planar manipulators: $Z \neq 0$ goals unreachable
- **Nearest Attainable Frame** problem becomes critical

$$\min_{\theta} \| {}^{0}_{N}T_{desired} - f(\theta) \|$$

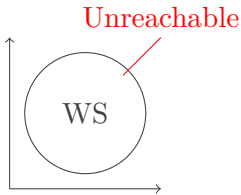
Tool Frame vs Wrist Frame

- Workspace computed for wrist frame $\{W\}$
- Tool frame $\{T\}$ transformations handled separately
- Computational check: Is $\{W\}$ pose feasible?

Practical Considerations

Implementation Challenges

- Joint limit handling
- Singularity avoidance
- Computational efficiency
- Multiple solution resolution



Key Questions

- Does solution exist? (WS check)
- How many solutions? (Configuration space)
- Which solution to choose? (Optimization)

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Section 2.1

Configuration Choice and Solution Space

Multiple Solutions Phenomenon

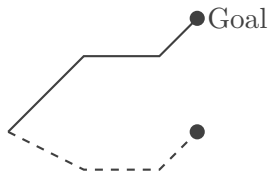
2 Multiple Solutions Analysis

Fundamental Challenge

Nonlinear kinematic equations often yield multiple valid joint configurations for a single end-effector pose

Example: 3-Link Planar Arm

- Large dextrous workspace
- Multiple configurations achieve same pose
- Dashed lines show alternative solution



Solution Selection Criteria

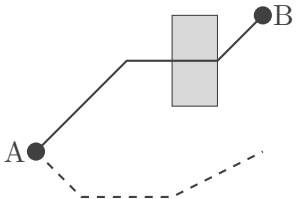
2 Multiple Solutions Analysis

Optimal Configuration Choice

- Minimize joint displacement
- Weighted joint movement

$$\min \sum w_i |\Delta\theta_i|$$

- Obstacle avoidance
- Energy efficiency



Critical Requirement

Must calculate *all* possible solutions for informed decision-making

PUMA 560 Case Study

2 Multiple Solutions Analysis

8 Potential Solutions

Four basic configurations with flipped wrist variants

Wrist Flip Transformation

$$\theta_4' = \theta_4 + 180^\circ$$
$$\theta_5' = -\theta_5$$
$$\theta_6' = \theta_6 + 180^\circ$$

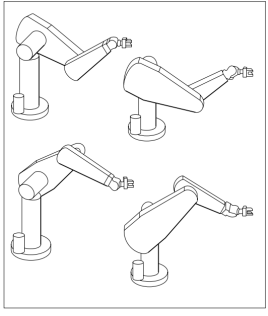


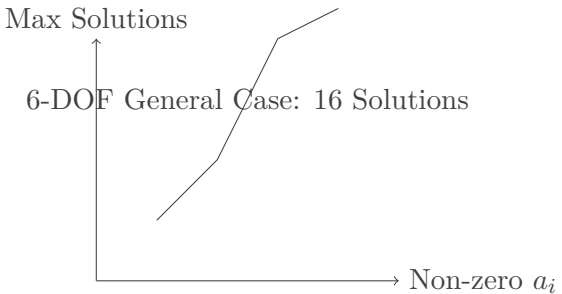
Figure 1: Enter Caption

Solution Space Dimensions

2 Multiple Solutions Analysis

Key Determinants

- Number of non-zero link parameters (a_i)
- Joint range limitations
- Manipulator architecture (rotary/prismatic)

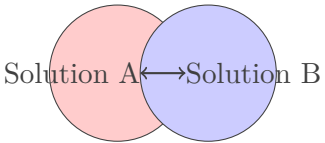


Computational Challenges

2 Multiple Solutions Analysis

Implementation Considerations

- Exhaustive solution enumeration
- Singularity detection
- Joint limit constraints
- Real-time performance
- Solution ranking metrics



Industrial Reality

Typical implementations precompute & cache solutions for critical poses

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Definition of Solvability

3 Methods of Solution for Inverse Kinematics

Formal Criterion

A manipulator is **solvable** if:

- All joint variables can be determined algorithmically
- *All* solution sets are computable for a given pose

Exclusion

Numerical iterative methods not guaranteed to find all solutions are excluded from this definition

Example

PUMA 560: 8 potential solutions via closed-form, but some may be inaccessible due to joint limits

Solution Method Taxonomy

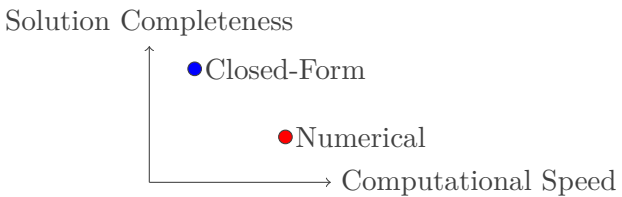
3 Methods of Solution for Inverse Kinematics

Closed-Form Solutions

- Analytic expressions
- Polynomials (degree ≤ 4)
- Non-iterative
- Subclasses:
  - Algebraic
  - Geometric

Numerical Solutions

- Iterative algorithms
- Convergence not guaranteed
- Computationally intensive
- Specialized applications:
  - Complex geometries
  - Singularity handling



Closed-Form Sufficient Conditions

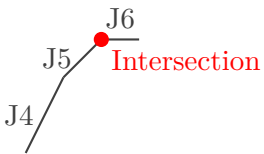
3 Methods of Solution for Inverse Kinematics

Theorem (Pieper's Criteria)

A 6-DOF serial manipulator with revolute joints admits closed-form solutions if:

- *Three consecutive joint axes intersect at a common point*
- ***OR*** *multiple* $\alpha_i = 0°$ *or* $\pm 90°$

Example (PUMA 560)



Design Implication

Modern industrial robots intentionally satisfy these conditions for real-time control

Algebraic vs Geometric Methods

Algebraic Approach

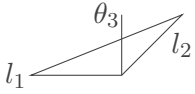
- Symbolic equation manipulation
- Trigonometric identities
- Example:

$$\theta_1 = \arctan\left(\frac{p_y}{p_x}\right) \pm \arccos\left(\frac{D}{2l_1l_2}\right)$$

Geometric Approach

- Law of cosines
- Projection decomposition
- Example:

$$c_3 = \frac{p_x^2 + p_y^2 - l_1^2 - l_2^2}{2l_1l_2}$$



Practical Implementation

3 Methods of Solution for Inverse Kinematics

Industrial Design Practices

- Strategic joint alignment (3-axis intersection)
- Simplified DH parameters ($\alpha_i = 0°/90°$)
- Precomputed solution libraries

Computation Pipeline

1. Check solution existence (WS)
2. Calculate all closed-form solutions
3. Filter by joint limits
4. Select optimal configuration



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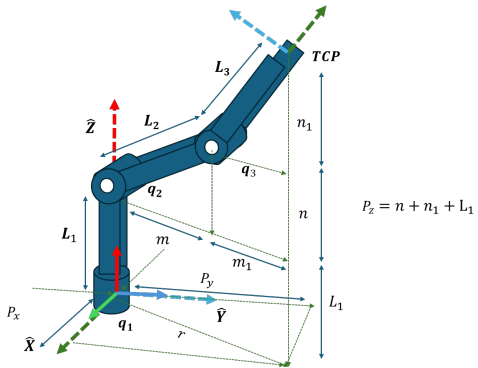
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Section 4.1

Angular Robot

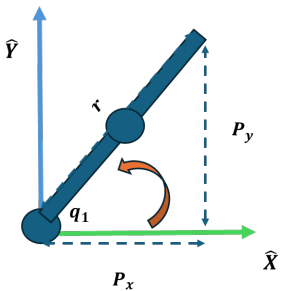
Figure 2: *Angular Robot 3GDL RRR.*



Inverse Kinematics solution

4 Inverse Kinematics Angular Robot

Figure 3: *Angular Robot 3GDL RRR top view.*



Geometric analysis

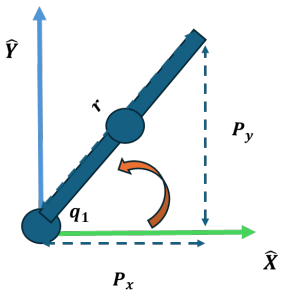
Pythagorean theorem

$$r^2 = P_x^2 + P_y^2$$

$$r = \sqrt{P_x^2 + P_y^2}$$

Inverse Kinematics solution get r

Figure 4: *Angular Robot 3GDL RRR top view.*



Geometric analysis

Pythagorean theorem

$$r^2 = P_x^2 + P_y^2$$

$$r = \sqrt{P_x^2 + P_y^2}$$

Inverse Kinematics solution get q_1

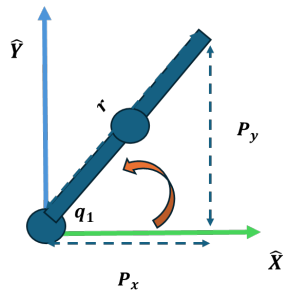
Geometric analysis

Tangent function

$$\tan q_1 = \frac{P_y}{P_x}$$

$$q_1 = \tan^{-1} \frac{P_y}{P_x}$$

Figure 5: *Angular Robot 3GDL RRR top view.*



Inverse Kinematics solution get q_2

Pythagorean theorem

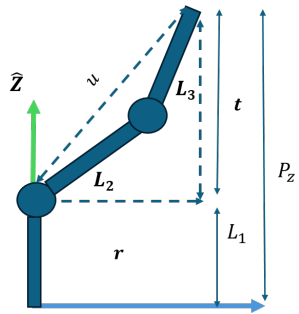
$$u^2 = t^2 + r^2$$

$$P_z = t + L_1$$

$$t = Pz - L_1$$

$$u = \sqrt{(Pz - L_1)^2 + (P_x^2 + P_y^2)}$$

Figure 6: *Angular Robot 3GDL RRR side view.*

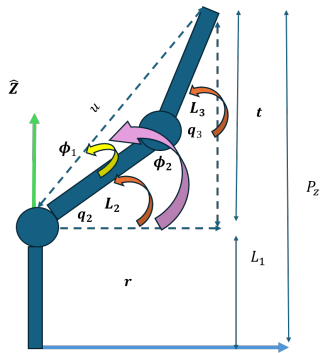


Inverse Kinematics solution get q_2

$$\phi_2 = \phi_1 + q_2$$

$$q_2 = \phi_2 - \phi_1$$

Figure 7: *Angular Robot 3GDL RRR side view.*



Inverse Kinematics solution get ϕ_2

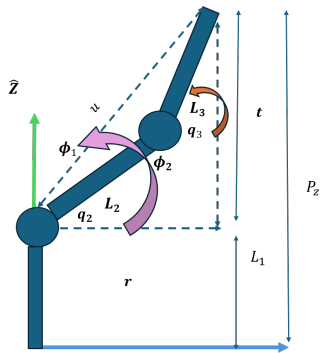
ϕ_2 is the angle between r and u

Tangent function

$$\tan \phi_2 = \left(\frac{t}{r} \right)$$

$$\phi_2 = \tan^{-1} \left(\frac{Pz - L_1}{\sqrt{P_x^2 + P_y^2}} \right)$$

Figure 8: *Angular Robot 3GDL RRR side view.*



Inverse Kinematics solution get ϕ_1

ϕ_1 is the angle between L_2 and u

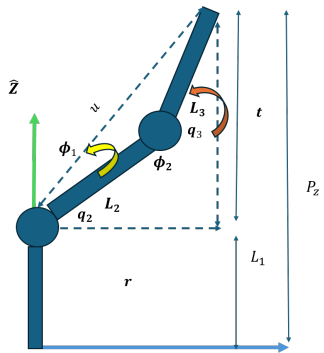
Law of cosines

$$L_3^2 = L_2^2 + u^2 - 2L_2 u \cos\phi_1$$

$$\phi_1 = \cos^{-1}\left(\frac{L_3^2 - L_2^2 - u^2}{-2L_2 u}\right)$$

$$\phi_1 = \tan^{-1}\left(\frac{\sqrt{1 - \left(\frac{L_3^2 - L_2^2 - u^2}{-2L_2 u}\right)^2}}{\frac{L_3^2 - L_2^2 - u^2}{-2L_2 u}}\right)$$

Figure 9: *Angular Robot 3GDL RRR side view.*



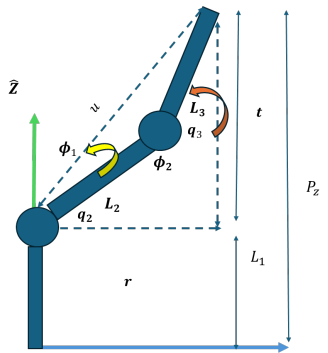
Inverse Kinematics solution get q_3

ϕ_3 is the angle between L_2 and L_3
A line is at 180 degrees when it forms a straight angle, that is, an angle that measures 180 degrees.

$$180 = \phi_3 + q_3$$

$$q_3 = 180 - \phi_3$$

Figure 10: *Angular Robot 3GDL RRR side view.*



Inverse Kinematics solution get q_3

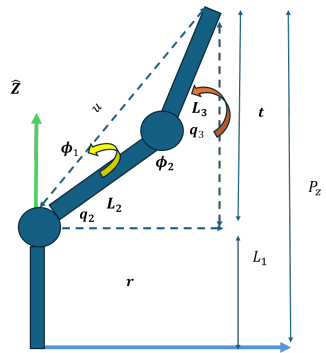
Law of cosines

$$u^2 = L_2^2 + L_3^2 - 2L_2L_3\cos\phi_3$$

$$\phi_3 = \cos^{-1}\left(\frac{u^2 - L_2^2 - L_3^2}{-2L_2L_3}\right)$$

$$\phi_3 = \tan^{-1}\left(\frac{\sqrt{1-\left(\frac{u^2-L_2^2-L_3^2}{-2L_2L_3}\right)^2}}{\frac{u^2-L_2^2-L_3^2}{-2L_2L_3}}\right)$$

Figure 11: *Angular Robot 3GDL RRR side view.*



Solution

4 Inverse Kinematics Angular Robot

$$q_1 = \tan^{-1}\left(\frac{P_y}{P_x}\right)$$

$$q_2 = \phi_2 - \phi_1$$

$$q_2 = \tan^{-1}\left(\frac{Pz - L_1}{\sqrt{P_x^2 + P_y^2}}\right) - \tan^{-1}\left(\frac{\sqrt{1 - \left(\frac{L_3^2 - L_2^2 - u^2}{-2L_2u}\right)^2}}{\frac{L_3^2 - L_2^2 - u^2}{-2L_2u}}\right)$$

$$q_3 = 180 - \phi_3$$

$$q_3 = 180 - \tan^{-1}\left(\frac{\sqrt{1 - \left(\frac{u^2 - L_2^2 - L_3^2}{-2L_2L_3}\right)^2}}{\frac{u^2 - L_2^2 - L_3^2}{-2L_2L_3}}\right)$$

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Q&A

Thank you for your attention

Rodriguez, J. Fundamentos De Robotica Antonio Barrientos.
Craig, J. J. (2006). Introduction to robotics. Pearson Educacion..
Spong, M. W., Hutchinson, S., & Vidyasagar, M. (2020). Robot modeling and control. John Wiley & Sons.